

Graphing Functions and Data with Desmos

Desmos Graphing Calculator · Desmos

Years 7-10	~30 minutes	Mathematics · Algebra & Functions
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NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.desmos.com/calculator>

Curriculum links: AC9M7A04 · AC9M7A05 · AC9M10A05 · AC9M10ST03

Learning intentions

- I can use Desmos to plot points and graph the rule of a function.
- I can add a slider to an equation and describe how changing its value changes the graph.
- I can build a scatterplot from a table of data and describe the association between two variables.

Getting started

Open the Desmos Graphing Calculator in your browser. Start with a blank graph and type an equation straight into the input list to see it plotted instantly — try this once just to see how it works before you start the tasks.

Tasks

1. Type $y = 2x + 3$ into Desmos. Record the coordinates of three points that lie on this line.

2. Rewrite the equation as $y = mx + 3$ and click the slider button that appears for m . Drag the slider and describe what happens to the line as m changes.

3. Set the slider to three different values of m (e.g. -2, 0, 3) and record the effect on the line each time.

Value of m	Effect on the line

4. Based on your table, explain in your own words what the number 'm' controls in $y = mx + c$, and what 'c' controls.

5. Add a table in Desmos and enter x-values 0 to 5 next to the rule $y = x^2$. Describe the shape made by the plotted points.

6. Enter this data as a table in Desmos: hours of study (1, 2, 3, 4, 5) against test score (example values you choose, e.g. increasing roughly with study time). Describe the pattern the resulting scatterplot shows.

Hours studied	Test score (your example values)

7. Describe the strength, direction and linearity of the association shown in your scatterplot.

8. Identify one point in your scatterplot that looks like an outlier, or explain why there isn't one. What might explain an outlier in real data like this?

Extension (optional)

Use a slider to build $y = a(x - h)^2 + k$ and experiment with all three parameters together. Explain how each one moves or reshapes the parabola compared to $y = x^2$.
